

RICHIAMO sul MOTO ONDULATORIO:

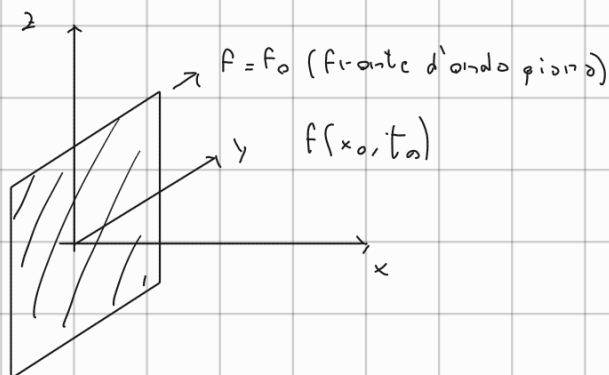
$$f(x, t) = A e^{i(\overbrace{kx \pm \omega t}^{\text{FASE}})}$$

$\downarrow$
 $A_{\text{imp.}}$

$\ominus \Rightarrow$ PROGRESS.

$$v = \frac{\omega}{k}$$

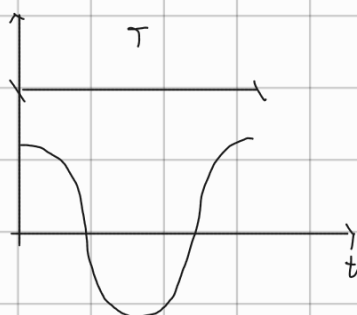
$\oplus \Rightarrow$ REGRESS.



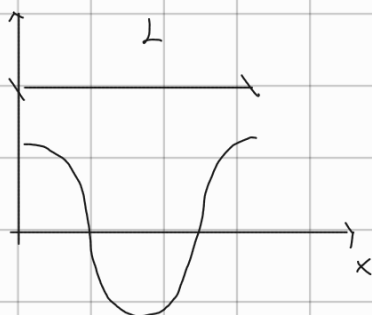
↑ FISSATA dallo SORVENTE

$\lambda v = |\vec{v}|$

$$\text{Re}\{f(x_0, t)\}$$



$$\text{Re}\{f(x, t_0)\}$$



• ONDE 3D:

$$f(\vec{r}, t) = A(r) e^{i(\vec{k} \cdot \vec{r} \pm \omega t)} \quad (\text{onda sferica})$$

$$A(r) = \frac{A_0}{r}$$

→ Onde Reali (lung. finita) → Pacchetto d'onda

$$f(x, t) = f_1(x, t) + f_2(x, t) = A \cos(k_1 x - \omega_1 t) + A \cos(k_2 x - \omega_2 t)$$

$$f(x, t) = A \cos(\Delta k x - \Delta \omega t) \cos(k_m x - \omega_m t)$$

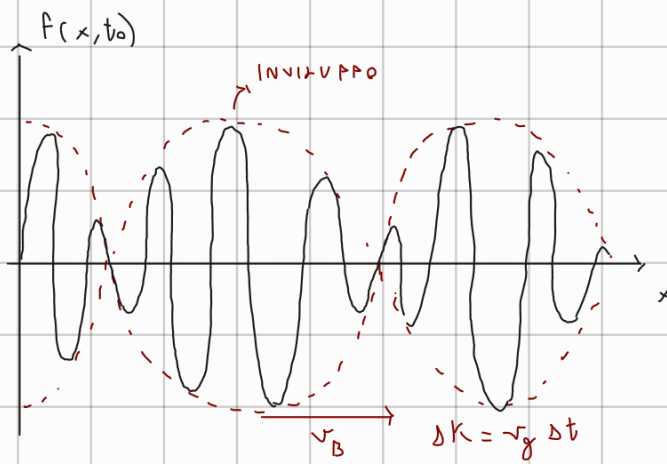
$$\Delta \omega = \frac{\omega_1 - \omega_2}{2}$$

$$\omega_m = \frac{\omega_1 + \omega_2}{2}$$

$$\Delta k = \frac{k_1 - k_2}{2}$$

$$k_m = \frac{k_2 + k_1}{2}$$

BATTIMENTI



$$v_f (\text{vel. di fase}) = \frac{\omega_n}{k_n}$$

$$v_g (\text{vel. di gruppo}) = \frac{\Delta \omega}{\Delta k}$$

• MEZZO $\begin{cases} \text{NON DISPERSIVO } v_f = v_g \\ \text{DISPERSIVO } v_f \neq v_g \end{cases}$

$$f(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} A(\omega) e^{i(Kx - \omega t)} d\omega$$

$\Rightarrow$ Somma di onde piane armoniche

$$f(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} A(k) e^{i(Kx - \omega t)} dk$$

ognuna con la sua ampiezza

$\longrightarrow$ Se il mezzo non assorbe: $A(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(x, 0) e^{-ikx} dx$

$\downarrow$ TRASFORMATA di FOURIER

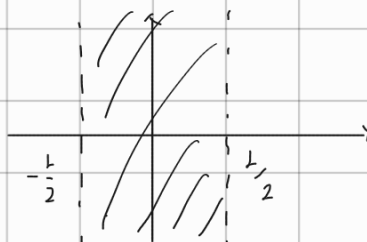
$$v_f = \frac{\omega_n}{k_n} \quad v_g = \lim_{\Delta k \rightarrow 0} \frac{\Delta \omega}{\Delta k} = \frac{d\omega}{dk}$$

$$\boxed{\omega = \omega(k)} \Rightarrow \text{RELAZIONE di DISPERSIONE}$$

$\downarrow$ Non disp. $\Rightarrow \omega = ck$

E.s.:

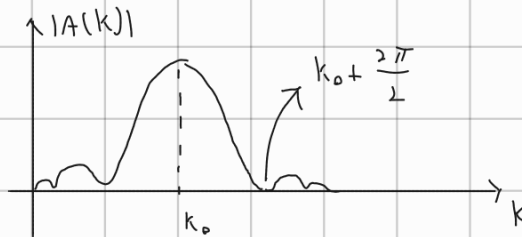
$$f(x, 0) = \begin{cases} c e^{ik_0 x} & -\frac{L}{2} \leq x \leq \frac{L}{2} \\ 0 & \text{Altrove} \end{cases}$$



$$A(k) = \frac{1}{\sqrt{2\pi}} \int_{-\frac{L}{2}}^{\frac{L}{2}} c e^{ik_0 x} e^{-ikx} dx = \frac{c}{\sqrt{2\pi}} \int_{-\frac{L}{2}}^{\frac{L}{2}} e^{i(k_0 - k)x} dx$$

$$= \frac{c}{\sqrt{2\pi}} \left[\frac{e^{i(k_0 - k)x}}{i(k_0 - k)} \right]_{-\frac{L}{2}}^{\frac{L}{2}} = \frac{c}{\sqrt{2\pi}} \left[\frac{e^{i(k_0 - k)\frac{L}{2}} - e^{-i(k_0 - k)\frac{L}{2}}}{i(k_0 - k)} \right]$$

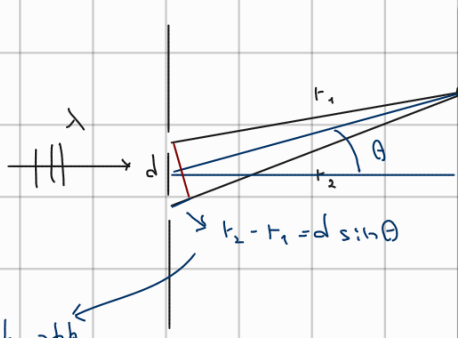
$$= \frac{2c}{\sqrt{2\pi}} \frac{\sin\left[(k_0 - k)\frac{L}{2}\right]}{k_0 - k} \quad (\text{sin card.})$$



se $L \rightarrow +\infty$ gli zeri del sinc si schiacciano su $K \rightarrow$ quindi: tornano ad una onda monocromatica (con un solo K)

al contr. più il pochi. dura poco più considera tante freq.

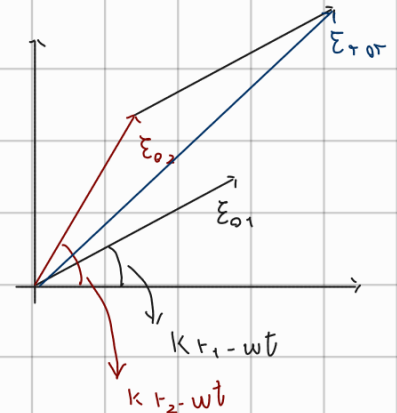
ESPER. di YOUNG:



$$E_1(P) = E_{01} e^{i(Kr_1 - \omega t)}$$

$$E_2(P) = E_{02} e^{i(Kr_2 - \omega t)}$$

$$E_{TOT}(P) = E_1(P) + E_2(P)$$



$$E_{TOT} = \sqrt{E_{01}^2 + E_{02}^2 + 2E_{01}E_{02}\cos\beta}$$

$$\beta = (Kr_2 - \omega t) - (Kr_1 - \omega t) = K(r_2 - r_1) \text{ (diff. di cammino)}$$

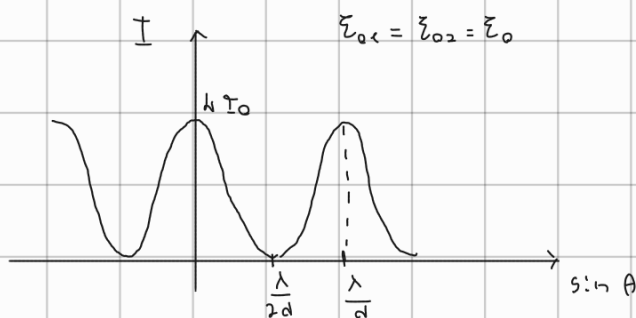
STAZIONARIO

$$I(P) \propto |E_{TOT}(P)|^2 = |E_1 + E_2|^2$$

$$I(P) = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(K(r_2 - r_1))$$

$$MAX \rightarrow K(r_2 - r_1) = 2n\pi \quad n = 0, \pm 1, \pm 2, \dots \Rightarrow Kd \sin \theta = 2n\pi \Rightarrow \sin \theta_{MAX} = n \frac{\lambda}{d}$$

$$MIN \rightarrow K(r_2 - r_1) = (2m+1)\pi \quad m = 0, \pm 1, \pm 2, \dots \Rightarrow \sin \theta_{MIN} = (2m+1) \frac{\lambda}{2d}$$



$$y_{MAX} = n \frac{\lambda L}{d}$$

• N SORGENTI:

$$I(\theta) = I_0 \left(\frac{\sin(N \frac{\Delta}{2})}{\sin(\frac{\Delta}{2})} \right)^2$$

• Max Ass: $\sin\left(\frac{\Delta}{2}\right) = 0 \rightarrow \frac{\Delta}{2} = m\pi \rightarrow \Delta = 2m\pi \rightarrow \sin \theta_m = m \frac{\lambda}{d}$

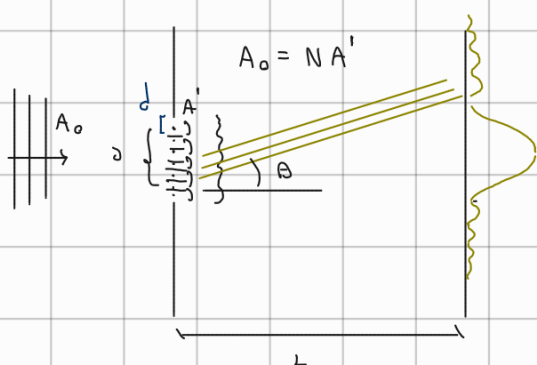
$\lim_{\Delta \rightarrow 0} I = \left(\frac{N \frac{\Delta}{2}}{\frac{\Delta}{2}} \right)^2 = N^2 \rightarrow \boxed{I = I_0 N^2}$

• Min Ass: $\sin\left(N \frac{\Delta}{2}\right) = 0 \rightarrow N \frac{\Delta}{2} = m\pi \rightarrow \Delta = 2Nm\pi \rightarrow \sin \theta_m = m \frac{\lambda}{Nd} \quad m \in \mathbb{Z}^0$

Ha due max avrà $N-1$ minimi con $I=0$ e avrà $N-2$ Max secondari

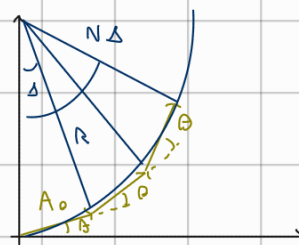
• Max REI: $\sin\left(N \frac{\Delta}{2}\right)^2 = 1 \rightarrow \sin(\theta'') = (2m''+1) \frac{\lambda}{2Nd} \quad m'' = 1, 2, \dots, N-2$

DIFFRAZIONE: $a = \lambda$



$Nd = a$
 $N \rightarrow +\infty$

$\Delta = Kd \sin \theta$

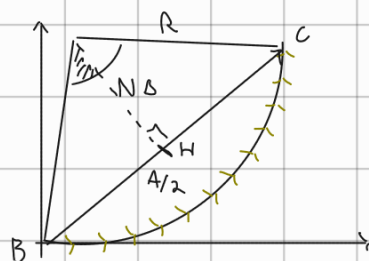


$\frac{A}{2} = R \sin\left(\frac{N\Delta}{2}\right)$

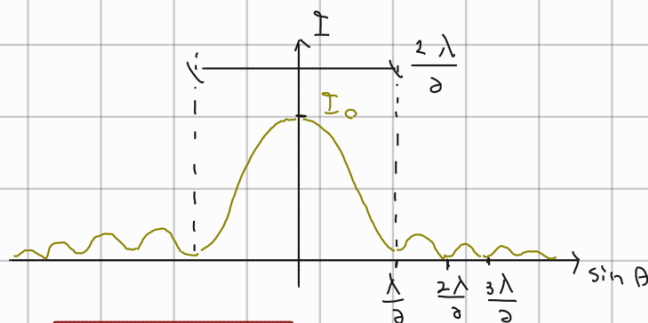
$BC = RND = NA'$

$\Rightarrow A = A_0 \text{sinc}\left(\frac{\pi a}{\lambda} \sin \theta\right)$

$\boxed{I = I_0 \text{sinc}^2\left(\frac{\pi a}{\lambda} \sin \theta\right)}$



$\text{sinc coordinate} \left(\frac{\sin(x)}{x} \right)$

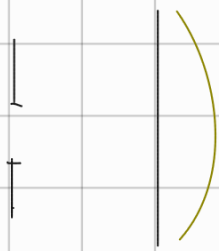


$\sin\left(\frac{\pi a}{\lambda} \sin \theta\right) = 0 \Rightarrow \frac{\pi a}{\lambda} \sin \theta = m\pi \Rightarrow \sin \theta_m = m \frac{\lambda}{a} \quad m \in \mathbb{Z}^0$

$\sin \theta_m \leq 1 \Rightarrow m \frac{\lambda}{a} \leq 1 \Rightarrow m \leq \frac{a}{\lambda}$

$m = \frac{a}{\lambda} \rightarrow \text{max ordine visibile se } a \gg \lambda \text{ lo spettro si suddivide}$

$$\lambda \approx a \Rightarrow \bar{m} = 1 \Rightarrow$$



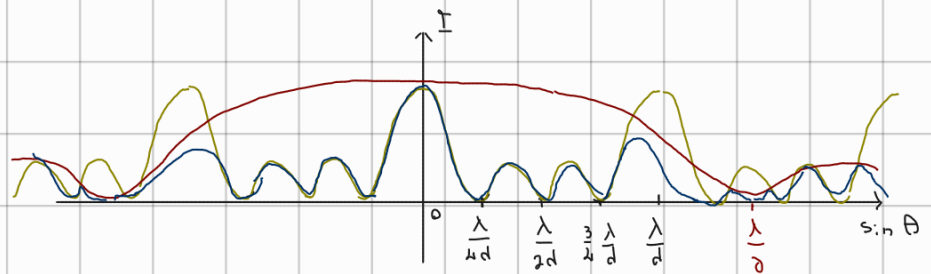
(Per praticamente tutte le appl. reali mi ritrovo ad avere uno certo grado di diffraz.)

RETICOLO di DIFFRAZIONE:

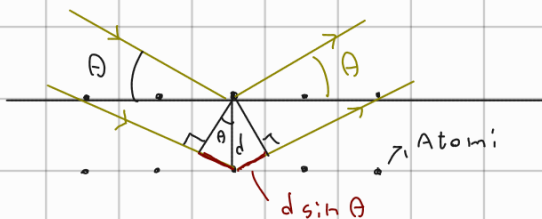
$$I = I_0 \left[\frac{\sin\left(\frac{\pi}{\lambda} a \sin \theta\right)}{\frac{\pi}{\lambda} a \sin \theta} \right]^2 \left[\frac{\sin\left(N \frac{\pi d}{\lambda} \sin \theta\right)}{\sin\left(\frac{\pi d}{\lambda} \sin \theta\right)} \right]^2$$

diff. interf.

$$N = 4$$



DIFFRA. di BRAGG:



$$\Delta = 2d \sin \theta$$

$$m \lambda \Rightarrow \Delta = 2m \lambda$$

$$\sin \theta_{\max} = m \frac{\lambda}{2d}$$

$$\approx 10^{-10} \text{ m} \Rightarrow \lambda = 10^{-12} \text{ m} \Rightarrow \text{ragg. X}$$

LENGO di MISURARE INS. ONDE e CORPUSCOLI

$$E = h \nu = \hbar \omega \quad N^{\circ} \text{ Fotoni: } \Rightarrow I$$

$$\vec{E}(k, t) = \vec{E} \cos(kx - \omega t)$$

ENERGIA FOTONE

$$E = \sqrt{p^2 c^2 + m^2 c^4} = \{m=0\}$$

$$\Rightarrow E = pc = h \nu$$

$$p = \frac{h \nu}{c} = \frac{h}{\lambda}$$

$$p = \frac{h}{2\pi} k = \hbar k \Rightarrow \boxed{\vec{p} = \hbar \vec{k}} \quad \text{Mom. del FOTONE}$$

$$I \propto |\vec{E}|^2 \Rightarrow N^{\circ} \text{ Fot.}$$

$$I = \mu_{em} c = \epsilon_0 |\vec{E}(x, t)|^2 c$$

$$\langle I \rangle = \frac{1}{2} \epsilon_0 \bar{E}^2 c =$$

$$\downarrow \quad \frac{W}{m^2} = \frac{J}{s \cdot m^2}$$

Fot. $\nearrow$ Flusso di foti
s. m² EN. di un fot [J]
 $\uparrow$ $\nearrow$
 $N \cdot \hbar \omega$

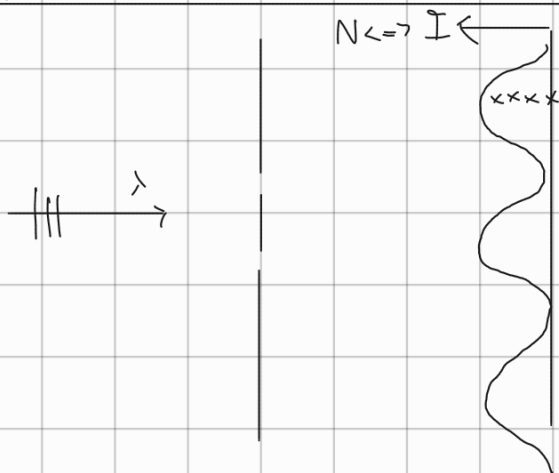
$$N = \frac{\epsilon_0 c \bar{E}^2}{2\pi \omega} = \frac{\langle I \rangle}{\hbar \omega}$$

Es. LASER $\hbar \omega = 2 \text{ eV (rosso)}$ $I = \frac{1 \text{ mW}}{\text{mm}^2} = \langle I \rangle = \frac{10^{-3} \text{ W}}{10^{-6} \text{ m}^2}$

$$N = \frac{\langle I \rangle}{\hbar \omega} = \frac{10^3 \text{ J/s m}^2}{2 \cdot 1,6 \cdot 10^{-19} \text{ J}} \approx 3 \cdot 10^{21} \frac{\text{fotoni}}{\text{s} \cdot \text{m}^2}$$

$$\bar{E} = \sqrt{\frac{N 2 \hbar \omega}{c \epsilon_0}} = 650 \frac{\text{V}}{\text{m}}$$

DUALISMO ONDA-PARTICELLA: FOTONI

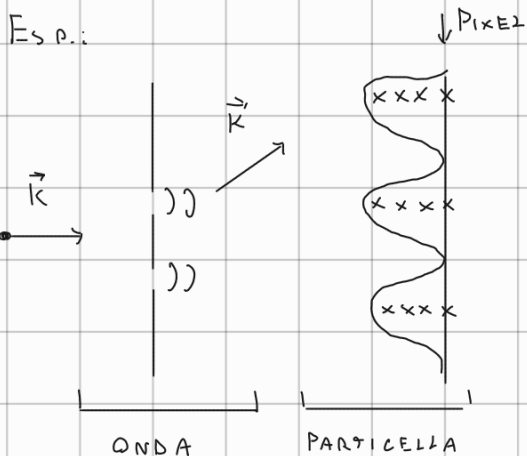


$$I = N \cdot \hbar \omega$$

$$N = \frac{I}{\hbar \omega}$$

$$I(\theta) = I_0 \left[\frac{\sin\left(\frac{2\pi d \sin\theta}{\lambda}\right)}{\sin\left(\frac{\pi d \sin\theta}{\lambda}\right)} \right]^2$$

$$N(\theta) = N_0 \cdot P(\theta)$$



MANDO 1 fot $1/\text{s m}^2$: pixel si accende/accende quando arriva un fotone. Dopo un po' compare la figura d'interf.

- Oss.: 1) Quando invio 1 fot. si accende un solo pixel $\Rightarrow$ il fotone lo misura sempre INTERO
- 2) Il fotone interferendo con le fend. deve acquisire nuove componenti di $\vec{K}$ (se no ANDREBBE DRITTO RISPETTO ALLE FENDITURE)

$\Rightarrow$ DEVE per forza interferire in qualche

modo, quindi deve comportarsi come

un'onda (che è definito contemporaneamente

in tutto lo spazio)

$\Rightarrow$ IL FOTONE INTERFERISCE CON SE STESSO

• LA LUCE È SIA ONDA CHE PARTICELLA

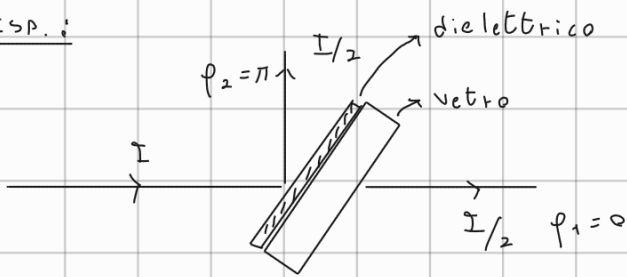
$\rightarrow$ MISURA $\Rightarrow$ PARTICELLA

$\rightarrow$ DURANTE IL TRAGITTO $\begin{cases} \text{ONDA (YOUNG)} \\ \text{PARTICELLA (COMPTON)} \end{cases}$

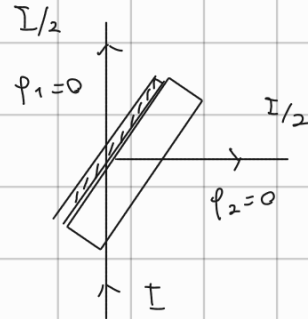
PRINCIPIO di COMPLEMENTARITÀ di BOHR:

→ Il tipo di esperimento determina il MODELLO (ondulatorio o corpuscolare) che deve essere per descrivere il fenomeno (TIPICAMENTE MUTUOESCLUSIVA)

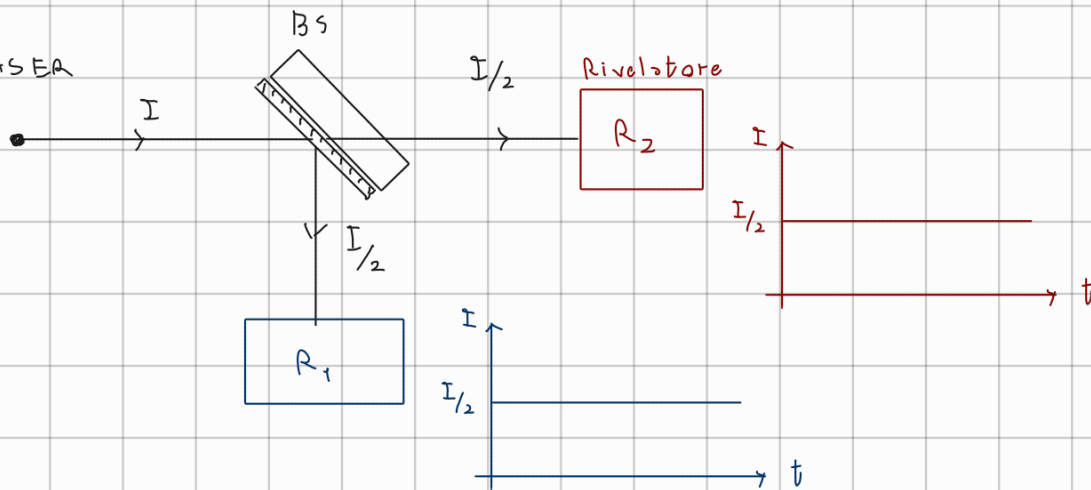
Esp.:



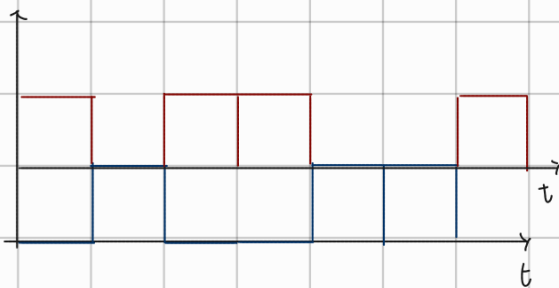
BEAM SPLITTER 50/50



1) LASER



ORA MANDO un fotone alla volta:



Non scottano mai insieme

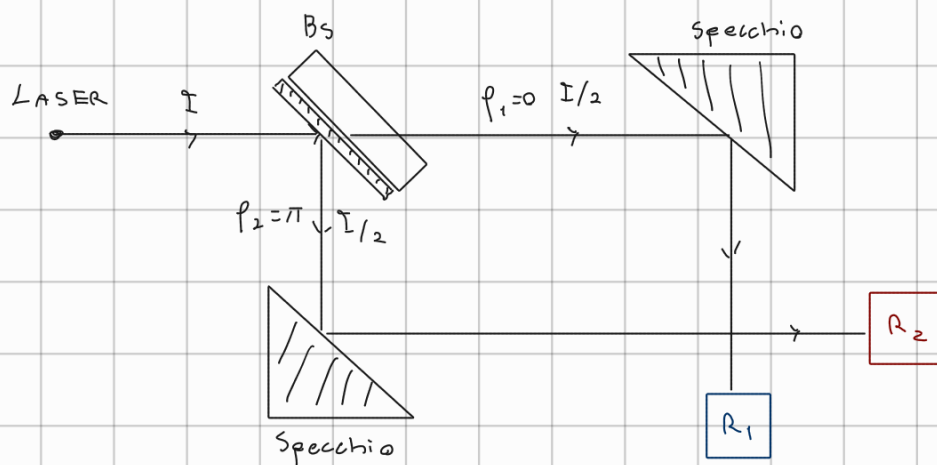
$$N^{\circ} \text{ conteggi } R_1 = N^{\circ} \text{ cont. } R_2$$

50% / 50%

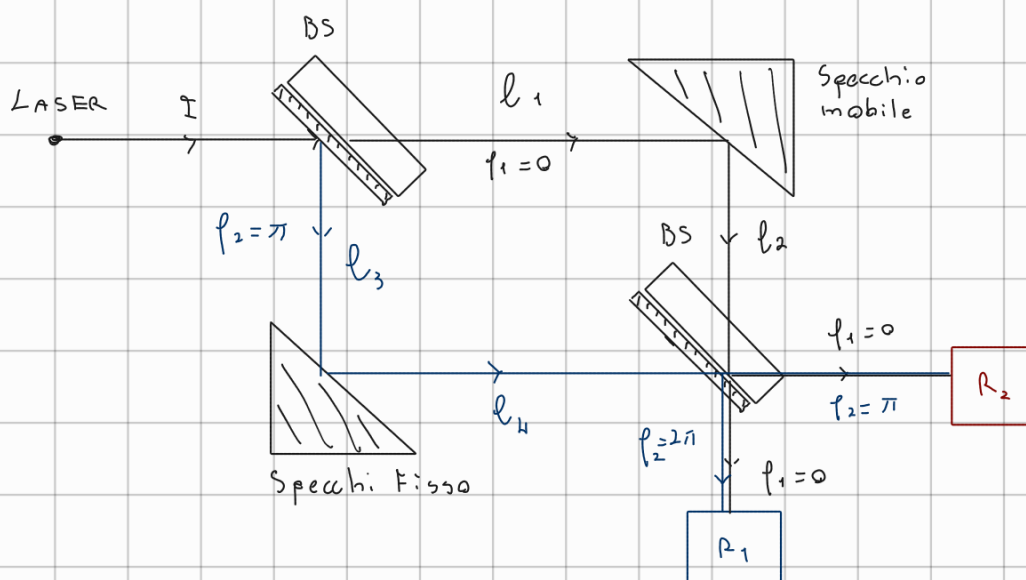
→ Il fotone lo misura sempre intero

→ " si comporta come una particella

2)



3) INTERF. di MACH-ZEHNDER

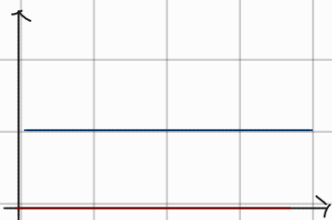


$$\boxed{R_1} \left. \begin{aligned} f_1^{\text{TOT}} &= K(l_1 + l_2) \\ f_2^{\text{TOT}} &= K(l_3 + l_4) + 2\pi \end{aligned} \right\} \Delta f_{R_1} = K(l_3 + l_4 - l_1 - l_2) + 2\pi$$

$$\boxed{R_2} \left. \begin{aligned} f_1^{\text{TOT}} &= K(l_1 + l_2) \\ f_2^{\text{TOT}} &= K(l_3 + l_4) + \pi \end{aligned} \right\} \Delta f_{R_2} = K(l_3 + l_4 - l_1 - l_2) + \pi$$

Ⓐ $K(l_3 + l_4 - l_1 - l_2) = 0 \Rightarrow \Delta f_{R_1} = 2\pi \rightarrow \text{INT. COSTRUTT.}$

$\Delta f_{R_2} = \pi \rightarrow \text{INT. DISTRUTT.}$



(B) $K(l_3 + l_4 - l_1 - l_2) = \pi \Rightarrow \Delta p_{R_1} = 3\pi \Rightarrow \text{INT. DISTR.}$

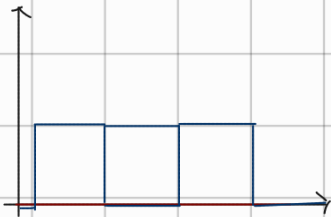
$\Delta p_{R_2} = 2\pi \Rightarrow \text{INT. COSTR.}$



• ORA INVIO SOLO 1 FOTONE

(A) $K(l_3 + l_4 - l_1 - l_2) = 0 \Rightarrow \Delta p_{R_1} = 2\pi \rightarrow \text{INT. COSTRUTT.}$

$\Delta p_{R_2} = \pi \rightarrow \text{INT. DISTRUTT.}$



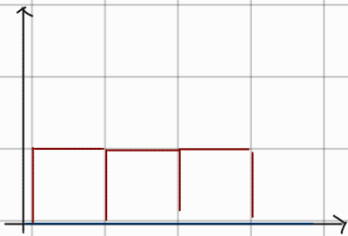
$\Rightarrow$ ARRIVANO TUTTI A R_1 , quando dovrai avere
suggera 50/50

$\Rightarrow$ Il fotone è delocalizzato, si trova in due
punti diversi nello stesso momento

$\Rightarrow$ SE BLOCCO UNA DELLE DUE STRADE TORNO A VEDERE 50/50

(B) $K(l_3 + l_4 - l_1 - l_2) = \pi \Rightarrow \Delta p_{R_1} = 3\pi \Rightarrow \text{INT. DISTR.}$

$\Delta p_{R_2} = 2\pi \Rightarrow \text{INT. COSTR.}$



• $|E_{TOT}|^2 = |E_1 + E_2|^2 \propto I$



$N(\theta) = N_0 P(\theta) \Rightarrow I \propto P \propto |E_1 + E_2|^2$

DUALISMO ONDA-PARTICELLA: ELETTRONI

1924: DE BROGLIE

→ FOTONE $\Rightarrow \Sigma(x,t) = \Sigma_0 e^{i(Kx - \omega t)}$

→ PARTICELLE $\Rightarrow \Psi(x,t) = A e^{i(Kx - \omega t)}$

→ Hp: $E = h\nu$ (di una partic.)

$p = \frac{h}{\lambda} \Rightarrow \boxed{\lambda = \frac{h}{p}}$ LUNG. D'ONDA di DE BROGLIE

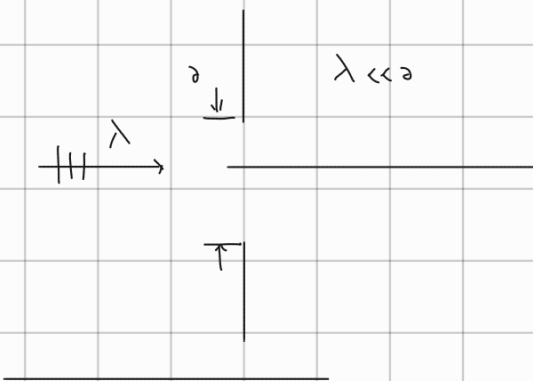
$\vec{p} = \hbar \vec{K} = \hbar K \vec{u}_{propagazione}$

Es.: $\left. \begin{array}{l} \text{Polvere} \\ m = 3 \cdot 10^{-9} \text{ kg} \\ \vec{v} = 2 \cdot 10^{-2} \text{ m/s} \end{array} \right\} \lambda = \frac{h}{p} \approx 10^{-23} \text{ m} \quad \lambda \rightarrow 0 \text{ (} 10^{-9} \text{ dimens. dell'atomo)}$

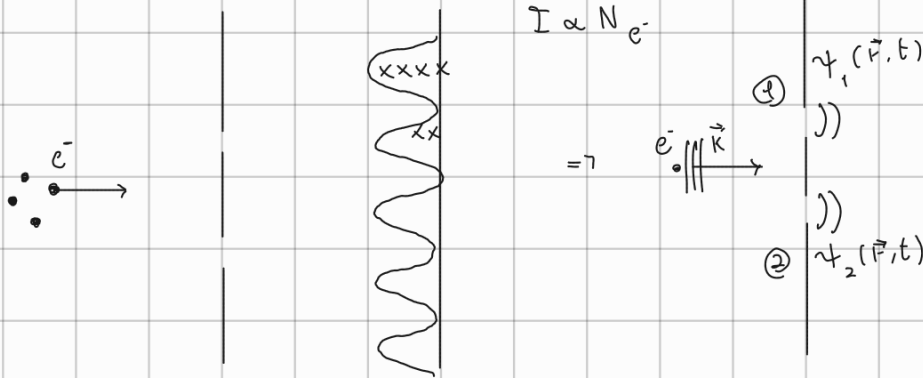
$|\vec{v}| \approx 10^{-11} \text{ m/s}$ (non esiste niente di così fermo) (agit. termica)

Es. 2:

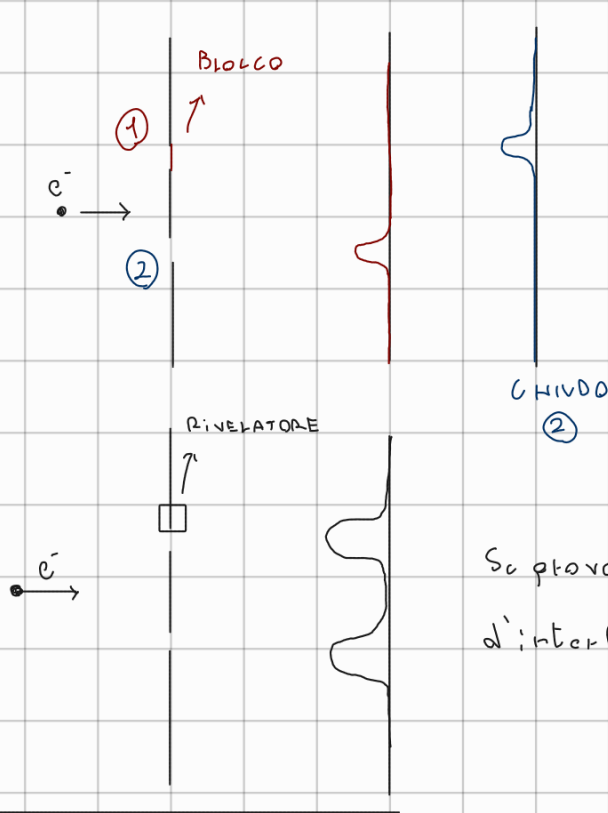
$e^- \rightarrow \vec{v} \quad K = 120 \text{ eV} = \frac{p^2}{2m} \quad \lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} \approx 10^{-10} \text{ m} = 1 \text{ \AA} \text{ (raggi X)}$
 $m_e = 9,1 \cdot 10^{-31} \text{ kg}$



→ ESPERIM. della doppia fend. con e^-



$\propto |\Psi_1 + \Psi_2|^2$
 è la quadratura
 componenti di K che
 prima non aveva
 DEVO ASSOCIARE ad e^-
 UNA FUNZIONE D'ONDA

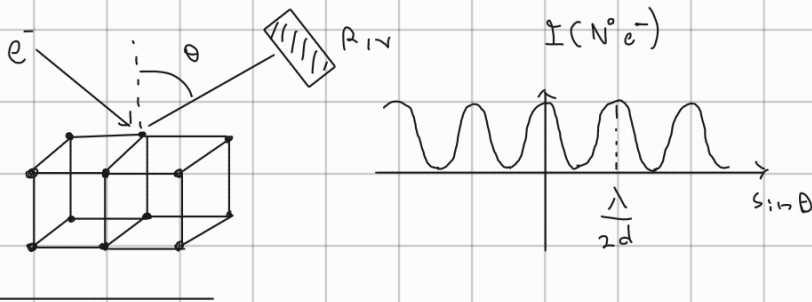


$$P \propto |\Psi|^2 = |\Psi_1 + \Psi_2|^2 \quad (\text{ONORE di MATERIA})$$

$$P \propto |\Sigma_{\text{TOT}}|^2 = |\Sigma_1 + \Sigma_2|^2$$

Se provo a vedere dove passo e^- distruggo la figura d'interferenza

DIFFRAZIONE di ELETTRONI:



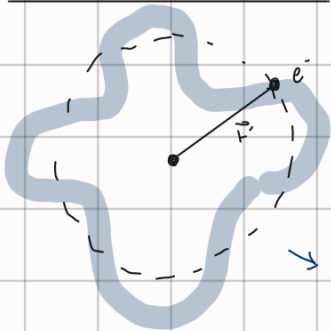
COME PER BRAGG

$$\sin \theta_{\max} = \frac{\lambda}{2d}$$

posso stimare λ di $e^- \approx 1 \text{ \AA}$

DUALISMO ONDA-PARTICELLA:

MODELLO di BOHR:



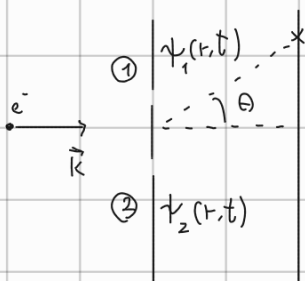
$$L = n\hbar = n \frac{h}{2\pi} = r p = r \frac{h}{\lambda} \Rightarrow \boxed{h\lambda = 2\pi r}$$

le uniche orbite permesse

$\Rightarrow$ Si comportano così le ONDE STAZIONARIE

$\rightarrow$ onda stazionaria ($h=h$)

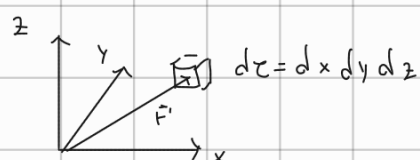
$\rightarrow$ Esp. due fenditure:



$$|\Psi|^2 = |\Psi_1 + \Psi_2|^2 \propto P$$

$$|\Psi(x,y,z,t)|^2 dx dy dz \rightarrow P \text{ di trovare un } e^- \text{ in } dx dy dz$$

all'istante t



$$P = |\Psi|^2 d\tau \quad \text{DENSITA' di PROBABILITA'}$$

$$\Rightarrow \int_{\mathbb{R}^3} |\Psi|^2 dx dy dz = 1$$

$$[|\Psi|^2] \rightarrow \frac{1}{m^3} \Rightarrow \Psi \text{ non ha significato fisico}$$

(B.E. ad esempio si misura Ψ NO)



$$\Psi(x, t) = A e^{i(Kx - \omega t)}$$

$$v_{\text{onda}} = \lambda v = \frac{h}{p} \cdot \frac{E}{h} = \frac{E}{p} = \frac{p^2}{2m} \cdot \frac{1}{p} = \frac{p}{2m} = \frac{m v_p}{2m} = \frac{v_p}{2}$$

Vel. partic.

$\Rightarrow$ UN'ONDA PIANA non è in grado di descriv. una particella

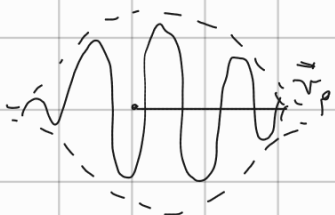
$$\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} A(k) e^{i(Kx - \omega t)} dk \quad (\text{PACCHETTO})$$

$$v_f = \frac{\omega}{k} = \frac{E/\hbar}{k}, \quad E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

$$\Rightarrow v_p = \frac{\hbar k^2}{2m} \cdot \frac{1}{k} = \frac{\hbar k}{2m} \Rightarrow \text{DISPERDE (ogni valore di } k \text{ ha vel. diverso)}$$

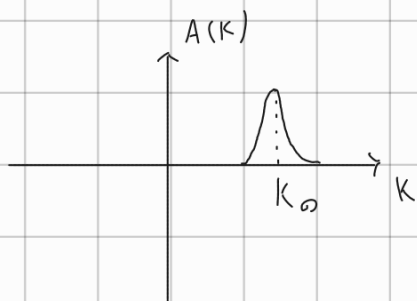
$$v_g = \frac{d\omega}{dk} = \frac{d(E/\hbar)}{d(p/\hbar)} = \frac{dE}{dp} = \frac{d}{dp} \left(\frac{p^2}{2m} \right) = \frac{p}{m} = \frac{m v_p}{m} = \underline{\underline{v_p}}$$

$$v_p = \frac{\omega}{k} \Rightarrow \omega = \frac{\hbar k^2}{2m} \quad \rightarrow \text{RELAZIONE DI DISPERSIONE}$$



$$\text{A 2 PROBAB. ONDA PIANA: } \Psi(x, t) = A e^{i(Kx - \omega t)}$$

$$\int_{\mathbb{R}} |\Psi|^2 dx dy dz \rightarrow +\infty \quad \text{DIVERGE}$$



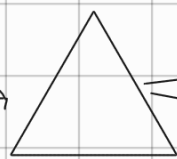
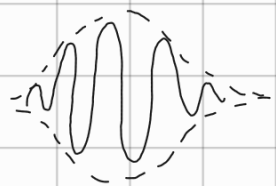
$\Rightarrow$ DA ORA IN POI LA NOSTRA ONDA PIANA È
COME UN PACCHETTO CON I k MOLTO VICINI A k_0 ,
QUASI COME SE IL RESTO SIA NULLO

• SIGNIFICATO FISICO di $A(k)$:

$$\Psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} A(k) e^{i(kx - \omega t)} dk \xrightarrow{\text{discretizzato}} \sum_k A(k) e^{i(kx - \omega t)} =$$

$$= \underbrace{A_1(k_1)}_{\text{NUMERO (VALORE IN CORRESP. di } k_1)} e^{i(k_1 x - \omega_1 t)} + A_2(k_2) e^{i(k_2 x - \omega_2 t)} + \dots$$

$$= \underbrace{\xi_{01}}_{A_1(k_1)} e^{i(k_1 x - \omega_1 t)} + \underbrace{\xi_{02}}_{A_2(k_2)} e^{i(k_2 x - \omega_2 t)} + \dots$$



PRISMA

ω_1

$$\xi_{01} e^{i(k_1 x - \omega_1 t)}$$

ω_2

$$\xi_{02} e^{i(k_2 x - \omega_2 t)}$$

$$N_1 \text{ fotoni} = \frac{c \epsilon_0 |\xi_{01}|^2}{2 \hbar \omega_1}$$

$$N_2 \text{ fot.} = \frac{c \epsilon_0 |\xi_{02}|^2}{2 \hbar \omega_2}$$

$$\Rightarrow |\xi_{01}|^2 = |A_1(k_1)|^2 \Rightarrow N_1 = \frac{c \epsilon_0 |A_1(k_1)|^2}{2 \hbar \omega_1}$$

$$N_2 = \frac{c \epsilon_0 |A_2(k_2)|^2}{2 \hbar \omega_2}$$

$\Rightarrow$ Più $A(k)$ è grande $\Rightarrow$ più
N. di fotoni con preciso k

$$\Rightarrow |A(k)|^2 \rightarrow P \text{ di avere un fotone con } k \div k + dk$$

Quindi: Fotoni $\rightarrow \underbrace{|A(k)|^2}_{\text{DENSITA' di PROBABILITA'}}$ $dk = P$ di avere un fotone con vett. d'onda

compreso tra $k \div k + dk$

DENSITA' di PROBABILITA'

$A(k) \Rightarrow$ Ampiezza di ogni comp. del pacchetto

• ELETTRONI (o PART.):

$$|A(k)|^2 dk = P \text{ di avere un } e^- \text{ con } k \div k + dk \text{ (} p \div p + dp \text{)}$$

$$|\Psi|^2 dz = P \text{ di trovare un } e^- \text{ in } dz$$

Relat. v. c (in qualsiasi
sist. di rif.
 $\uparrow$
il fot. viaggia
a c)

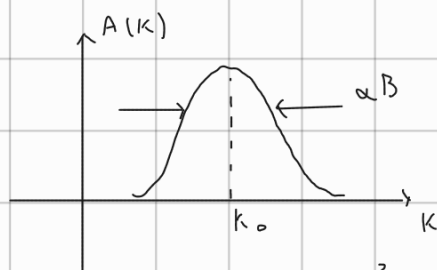
Elett.: $|\Psi|^2 dz$
 $|A|^2 dk$

Fot.: $|\xi|^2 dz \rightarrow P$ di trov. un fot. in dz ? $\Rightarrow$ NO
 $|A|^2 dk \rightarrow A(k)$ è la funz. d'onda del fotone

PRINCIPIO DI INDETERMINAZIONE DI HEISENBERG:

$$p_0 = \hbar k_0$$

$$A(k) = c e^{-\frac{(k-k_0)^2}{2B^2}}$$



$$\Psi(x,0) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} A(k) e^{ikx} dk = c' \int_{\mathbb{R}} e^{-\frac{(k-k_0)^2}{2B^2}} e^{ikx} dk \quad \begin{cases} q = k - k_0 \\ dq = dk \end{cases}$$

$$\Rightarrow c' \int_{\mathbb{R}} e^{-\frac{q^2}{2B^2}} e^{i(q+k_0)x} dq = c' e^{ik_0x} \int_{\mathbb{R}} e^{-\frac{q^2}{2B^2}} e^{iqx} dq =$$

$$= c' e^{ik_0x} \int_{\mathbb{R}} e^{-\left[\frac{q^2}{2B^2} - iqx\right]} dq$$

$$\Rightarrow \frac{q^2}{2B^2} - iqx = \frac{q^2}{2B^2} - iqx - \frac{x^2 B^2}{2} + \frac{x^2 B^2}{2} = \left[\frac{q}{\sqrt{2}B} - \frac{ixB}{\sqrt{2}} \right]^2 - \frac{x^2 B^2}{2}$$

$$\Rightarrow c' e^{ik_0x} \int_{\mathbb{R}} e^{-\left(\frac{q}{\sqrt{2}B} - \frac{ixB}{\sqrt{2}}\right)^2 - \frac{x^2 B^2}{2}} dq \quad \begin{cases} \mu = \frac{q}{\sqrt{2}B} - \frac{ixB}{\sqrt{2}} \\ dq = \sqrt{2}B d\mu \end{cases}$$

$$\Rightarrow c' e^{ik_0x} e^{-\frac{x^2 B^2}{2}} \int_{\mathbb{R}} \sqrt{2}B e^{-\mu^2} d\mu =$$

$$= \underbrace{\sqrt{2\pi} B c'}_{c''} e^{ik_0x} e^{-\frac{x^2 B^2}{2}}$$

$$\Psi(x,0) = c'' e^{ik_0x} e^{-\frac{x^2}{2 \frac{1}{B^2}}}$$

$$\Rightarrow |A(k)|^2 = |c|^2 e^{-\frac{(k-k_0)^2}{B^2}} \quad \begin{matrix} \text{Varianza} \\ \uparrow \end{matrix} \quad \Rightarrow 2\sigma_k^2 = B^2 \quad \begin{matrix} \text{dev. std.} \\ \uparrow \end{matrix} \quad \Rightarrow \sigma_k = \frac{B}{\sqrt{2}}$$

$$\Rightarrow |\Psi(x,0)|^2 = |c''|^2 e^{-\frac{x^2}{1/B^2}} \quad \Rightarrow 2\sigma_x^2 = \frac{1}{B^2} \quad \Rightarrow \sigma_x = \frac{1}{\sqrt{2}B}$$

$$\Rightarrow \sigma_k \sigma_x = \frac{1}{2}$$

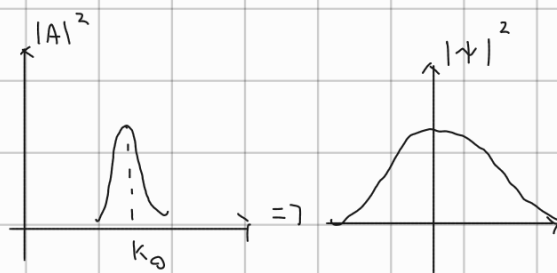
$\Rightarrow$ Se "allargato" $A(k)$ "stirato" Ψ

• $\sigma_k = \Delta k \rightarrow$ INCERTEZZA

• $\sigma_x = \Delta x$ "

$\Rightarrow p = \hbar k \Rightarrow \Delta p (\text{inc. sup}) = \hbar \Delta k$

$\Rightarrow \Delta k \Delta x = \frac{1}{2} \Rightarrow \Delta p \Delta x = \frac{\hbar}{2}$



$$\text{SE } t=0 \quad \Delta p \Delta x = \frac{\hbar}{2} ; \quad \Delta k \Delta x = \frac{1}{2}$$

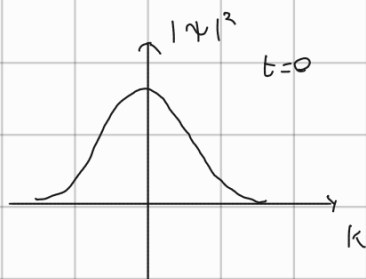
t > 0

$$\omega t = \frac{Et}{\hbar} = \frac{1}{\hbar} \underbrace{\hbar^2 k^2}_{E} t$$

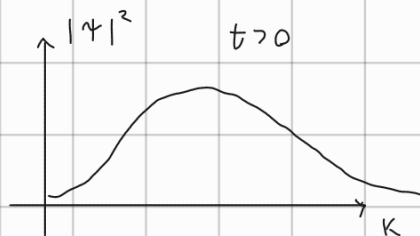
$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} A(k) e^{i(kx - \frac{\hbar k^2}{2m} t)} dk$$

$$\Rightarrow \sigma_x = \sigma_0 [1 + \rho(t)]$$

MONOTONA CRESCENTE



=>



$$\text{PACCHETTO GAUSSIANO: } \sigma_x \sigma_k = \frac{1}{2} \Rightarrow \Delta k \Delta x = \frac{1}{2} \Rightarrow \Delta p \Delta x = \frac{\hbar}{2}$$

$$f(x, t) \rightarrow f(x) = f(x, 0)$$

$$\rightarrow f(t) = f(0, t)$$

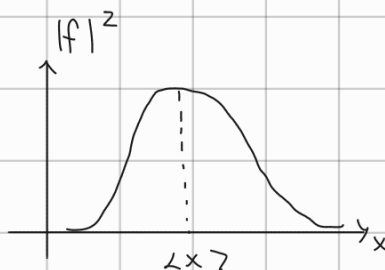
$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} A(k) e^{ikx} dk, \quad A(k) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(x) e^{-ikx} dx \Leftrightarrow$$

$$\Leftrightarrow f(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} G(\omega) e^{i\omega t} d\omega$$

$$G(\omega) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(t) e^{-i\omega t} dt$$

DEFINIZIONE CENTRO della $f(x)$: MEDIA

$$\langle x \rangle = \frac{\int_{\mathbb{R}} x |f(x)|^2 dx}{\int_{\mathbb{R}} |f(x)|^2 dx}$$



$$\langle k \rangle = \frac{\int_{\mathbb{R}} k |A(k)|^2 dk}{\int_{\mathbb{R}} |A(k)|^2 dk}$$

$$\Delta x = \sqrt{\frac{\int_{\mathbb{R}} (x - \langle x \rangle)^2 |f(x)|^2 dx}{\int_{\mathbb{R}} |f(x)|^2 dx}}$$

STESSA COSA PER Δk

$$\Rightarrow \Delta x \Delta K \geq \frac{1}{2}, \quad \Delta w \Delta t \geq \frac{1}{2} \Rightarrow \text{Per qualsiasi } f$$

Se la distr. è GAUSSIANA: $\Delta x \Delta k = \frac{1}{2}$, $\Delta \omega \Delta t = \frac{1}{2}$

CAUCHY-SCHWARTZ:

$$\vec{m}, \vec{r} \quad |\vec{m}, \vec{r}| \leq \|\vec{m}\| \cdot \|\vec{r}\| \rightarrow \text{Applico a } \Delta x \text{ e } \Delta r \text{ e trovo la dis. subito sopra}$$

CONSEGUENZA DIRETTA DEL FORMALISMO DI FOURIER (Trasformati)

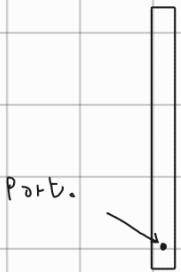
$$K = \frac{p}{\hbar}, \quad w = \frac{E}{\hbar} \Rightarrow \Delta x \Delta p \geq \frac{\hbar}{2} \quad \Delta E \Delta t \geq \frac{\hbar}{2} \Rightarrow \text{PRINCIPIO DI (INDETERMINAZIONE) DI HEISENBERG.}$$

Se como a condição. $x \Rightarrow \Delta x$ é 0

→ Größere stoch. bed. definite $\Rightarrow$ DEVI. STOR. $\Rightarrow$ INCREASE

→ Per ottenere dev. stndr devo fare tutte misure $\Rightarrow$ ENSEMBLE

$$\Rightarrow 2N \text{ syst. total: } \left. \begin{array}{l} \text{ugly: } \rightarrow \text{measure } N \text{ volts } x \Rightarrow \Delta x \\ \rightarrow \text{measure } N \text{ volts } v \Rightarrow p \Rightarrow \Delta p \end{array} \right\} \Delta x \Delta p \geq \frac{h}{2}$$



2N scatole uguali

N volte moltiplo $\times \Rightarrow \Delta x$ piccolo (simbolo stretto)

N sollte immer $\Rightarrow \Delta p$ GRASSE

→ ORIGINE FISICA?

Mikroskopie di Bohr - Heisenberg

Per capire dove sto e devo mandargli conto un fat.

=> MISURARE $\Rightarrow$ PERTURBARE il SISTEMA

